

Wind Farm Layout Optimization

Using Aeroacoustic Surrogate Modeling and OpenFAST Validation

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Abstract

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1 Introduction

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1.1 Motivation

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1.2 Research Objective

The primary objective of this work is to investigate whether a computationally efficient surrogate-based optimization framework can be used to optimize wind farm layouts subject to both power production and aeroacoustic constraints.

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2 Background

2.1 Wind Farm Layout Optimization

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2.2 FLORIS

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2.3 OpenFAST and FAST.Farm

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2.4 Aeroacoustic Modeling

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2.5 Genetic Algorithms

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3 Methodology

3.1 Overall Workflow

The optimization framework developed in this study consists of two key components: FLORIS and a custom machine-learning predictor for the steady-state aeroacoustic output of each turbine at a given receiver location. These computationally efficient approximators avoid the time and computational costs associated with repeatedly running OpenFAST, making them suitable for use within the genetic algorithm. However, they are not intended to serve as a complete replacement for a higher-fidelity simulation. For this reason, a final OpenFAST/FAST.Farm validation step is included.

During the optimization stage, a genetic algorithm evaluates candidate turbine layouts using FLORIS and the machine-learning aeroacoustic surrogate model. On the first iteration, candidates,

also referred to as chromosomes, are generated from a uniform random distribution within the constraints of the land plot. FLORIS is used to estimate wake interactions, effective turbine wind speeds at the hub, and overall farm power production for each candidate. The effective turbine wind speeds are then supplied to the aeroacoustic surrogate along with the candidate layouts to estimate the A-weighted sound level at each receiver per turbine per candidate.

The outputs of these models are incorporated into the genetic algorithm by providing information to calculate candidate fitness. The fitness function takes into account power production, as well as turbine-spacing, farm-boundary, and aeroacoustic constraints to output a single float metric. To evaluate candidates under a range of operating conditions, each generation is evaluated across a set of weighted wind-speed and wind-direction scenarios.

After each generation, chromosomes are selected based on fitness and used to produce the subsequent generation through crossover and random mutation. A portion of the lowest-performing population is replaced with newly generated random candidates to maintain population diversity, while the highest-performing candidate is retained through elitism. The specific genetic operators and optimization parameters are described in greater detail in the following sections.

Once the genetic algorithm reaches its termination criterion, currently defined as a fixed number of generations, the highest-performing candidate is evaluated using OpenFAST/FAST.Farm. The resulting higher-fidelity aeroacoustic predictions are then used to recalculate the candidate’s acoustic performance and fitness and to evaluate the agreement between the surrogate-based optimization and the OpenFAST validation.

3.2 Aeroacoustic Surrogate Model

3.2.1 Aeroacoustic Training Data

The training data for the aeroacoustic surrogate were generated using single-turbine OpenFAST simulations. Because each turbine was simulated independently, wake interactions between turbines were not included in the training-data generation process. Wake effects are instead accounted for later in the optimization workflow using FLORIS, which provides the effective wind speed experienced by each turbine.

Two primary variables were varied when generating the aeroacoustic training data: turbine wind speed and the position of the acoustic observer relative to the turbine. The wind-speed range was sampled from 3.0 to 24.5 m/s in increments of 0.5 m/s,

$$v \in \{3.0, 3.5, 4.0, \dots, 24.5\}, \quad (1)$$

resulting in 44 distinct wind-speed conditions.

Observer locations were defined in polar coordinates using radial distance r and azimuth angle θ . Twenty radial distances were generated using logarithmic spacing between 100 and 1000 m. A reference radius of 200 m was additionally included to provide a higher-resolution angular sampling at a representative observer distance. Each of the logarithmically spaced radii was evaluated at 16 azimuth angles, while the 200 m reference radius was evaluated at 32 azimuth angles.

The resulting observer sampling therefore contained

$$(20 \times 16) + 32 = 352 \quad (2)$$

observer positions. Combining these positions with the 44 wind-speed conditions produced a total of

$$N_{\text{samples}} = 44 \times 352 = 15,488 \quad (3)$$

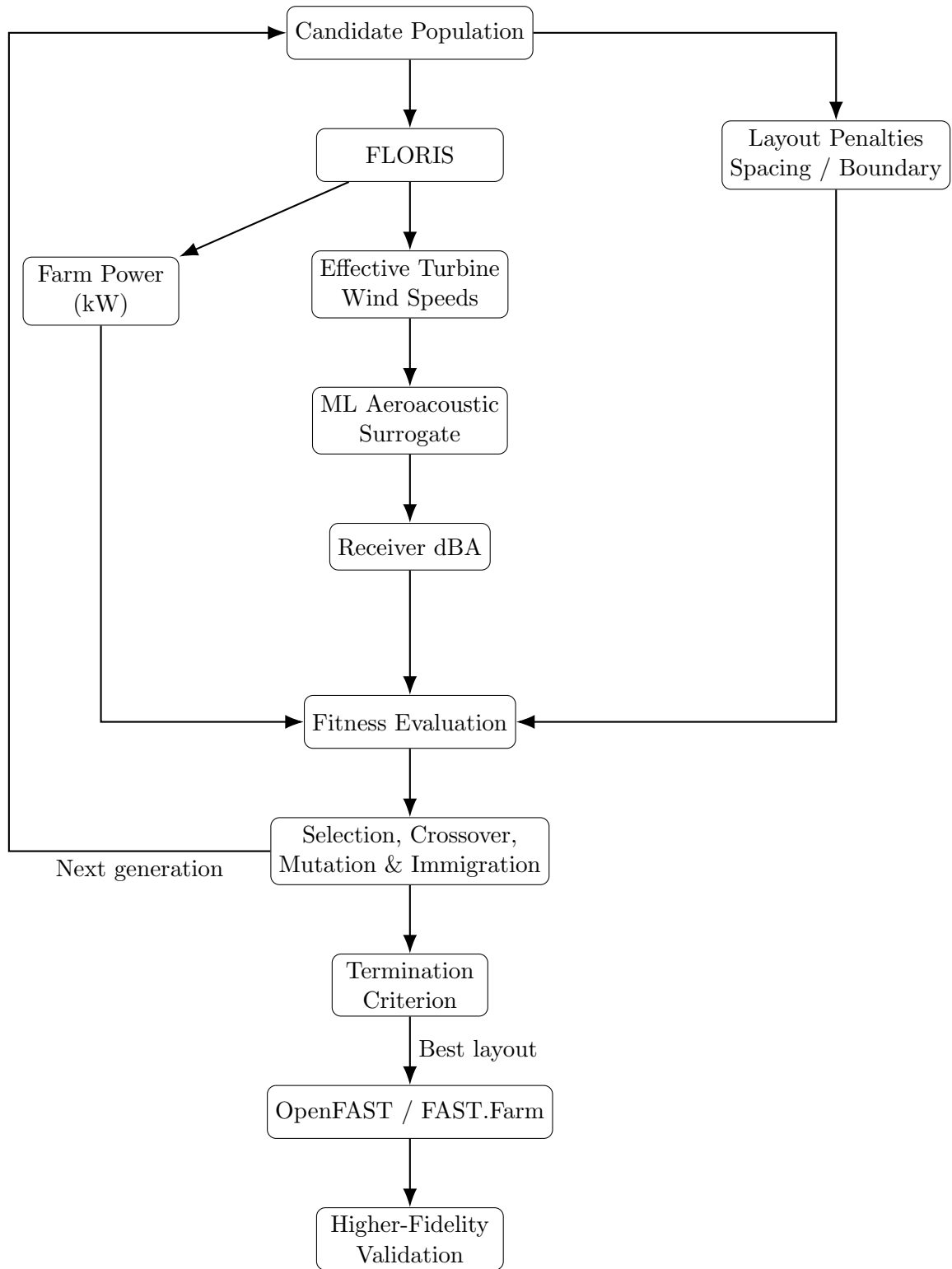
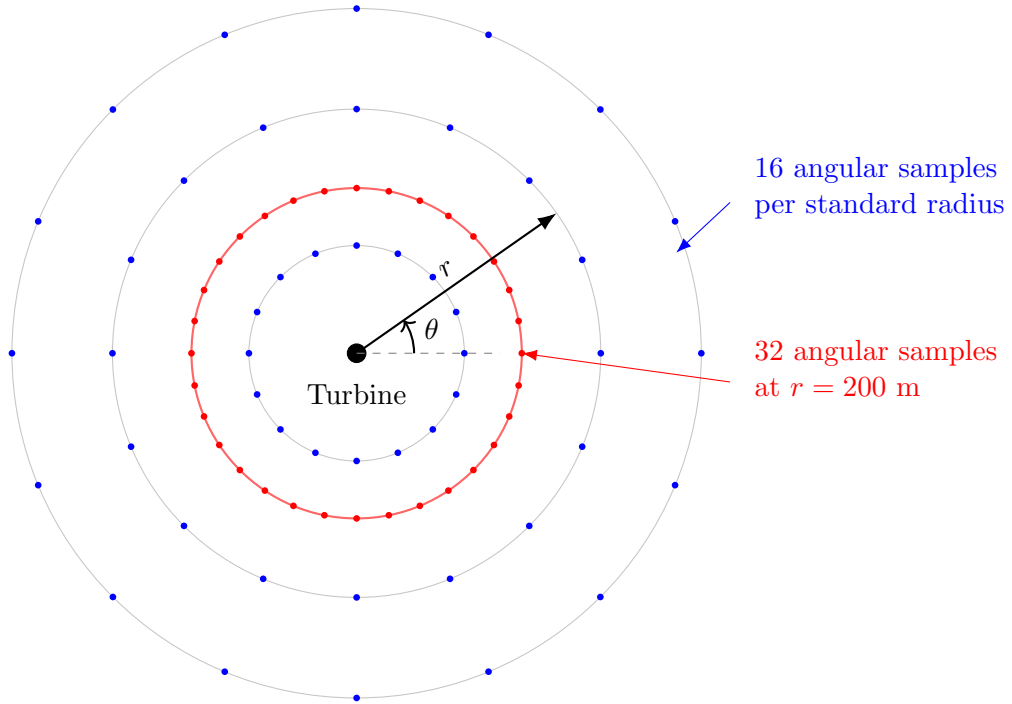


Figure 1: Overall genetic-algorithm optimization and OpenFAST/FAST.Farm validation workflow.

single-turbine aeroacoustic training samples.
Table 1 summarizes the sampling strategy.

Table 1: OpenFAST aeroacoustic training-data sampling.

Variable	Sampling	Number of Values
Wind speed v	3.0–24.5 m/s, 0.5 m/s increments	44
Radial distance r	Logarithmically spaced, 100–1000 m	20
Azimuth θ	16 angles per logarithmically spaced radius	16
Reference radius	200 m with 32 azimuth angles	32 observer positions
Total observer positions	$(20 \times 16) + 32$	352
Total training samples	44×352	15,488



Schematic only: radial spacing is logarithmic from 100–1000 m

Figure 2: Schematic representation of the observer sampling used to generate the single-turbine aeroacoustic training dataset. Standard radial distances use 16 uniformly distributed azimuth angles, while the 200 m reference radius uses a denser 32-angle sampling.

3.2.2 Candidate Feature Representations

The data generated from the OpenFAST simulations initially contained three input variables: wind speed v , relative observer coordinate X , and relative observer coordinate Y . Although the Cartesian coordinates fully describe the observer location, the aeroacoustic field exhibits a strong radial and directional structure. Several alternative feature representations were therefore evaluated to make this underlying geometry more explicit to the machine-learning models.

The relative Cartesian coordinates were first transformed into polar coordinates according to

$$r = \sqrt{X^2 + Y^2}, \quad (4)$$

and

$$\theta = \text{atan2}(Y, X). \quad (5)$$

Directly using θ as a numerical feature introduces an artificial discontinuity at the angular boundary. For example, directions immediately below 2π and immediately above 0 are physically adjacent but numerically distant. To preserve the periodic nature of the observer azimuth, the angular coordinate was therefore encoded using

$$c_\theta = \cos(\theta), \quad s_\theta = \sin(\theta). \quad (6)$$

Distance was also represented using several transformations. Under geometric spreading, acoustic intensity decreases with increasing distance, while sound-pressure level is expressed on a logarithmic scale. A logarithmic transformation of the observer distance,

$$r_{\log} = \log_{10}(r), \quad (7)$$

was therefore evaluated as a candidate feature. Reciprocal distance terms were additionally considered,

$$r^{-1} = \frac{1}{r}, \quad r^{-2} = \frac{1}{r^2}, \quad (8)$$

to determine whether explicitly representing distance-decay relationships improved predictive performance.

Rather than selecting one representation a priori, five candidate feature sets were constructed and evaluated with each candidate machine-learning model. These feature sets are summarized in Table 2. The complete set of original and engineered features is retained in `training_data.csv`.

Table 2: Candidate feature representations evaluated for the aeroacoustic surrogate.

Feature Set	Input Features	Purpose
Cartesian	v, X, Y	Original spatial representation
Polar	$v, r, \cos \theta, \sin \theta$	Explicit radial and directional representation
Polar + log distance	$v, \log_{10}(r), \cos \theta, \sin \theta$	Logarithmic distance representation
Polar + raw/log distance	$v, r, \log_{10}(r), \cos \theta, \sin \theta$	Combined raw and logarithmic distance
Polar + distance decay	$v, \log_{10}(r), r^{-1}, r^{-2}, \cos \theta, \sin \theta$	Multiple representations of distance decay

The feature representation was not selected independently of the regression model. Instead, each candidate machine-learning algorithm was trained and evaluated using all five feature sets. The final combination of model architecture and feature representation was selected based on predictive performance, as described in the following section.

3.2.3 Model Training and Selection

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3.2.4 Final Surrogate Implementation

The machine-learning surrogate uses the input feature vector

$$\mathbf{x} = [v \quad \log_{10}(r) \quad \cos(\theta) \quad \sin(\theta)] . \quad (9)$$

The surrogate then approximates the relationship

$$\hat{L}_A = f(v, \log_{10}(r), \cos(\theta), \sin(\theta)) , \quad (10)$$

where $\hat{L}_A$ is the predicted A-weighted sound level.

3.2.5 Aeroacoustic Aggregation

Independent turbine noise contributions are combined in the linear acoustic-energy domain:

$$L_{\text{total}} = 10 \log_{10} \left(\sum_{i=1}^N 10^{L_i/10} \right) . \quad (11)$$

Time-varying sound levels may similarly be reduced using

$$L_{\text{eq}} = 10 \log_{10} \left(\frac{1}{T} \sum_{t=1}^T 10^{L_t/10} \right) . \quad (12)$$

3.3 Genetic Algorithm

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3.3.1 Chromosome Representation and Initialization

3.3.2 Fitness Function

The optimization objective can be expressed generally as

$$F = P_{\text{expected}} - C_{\text{spacing}} - C_{\text{noise}} - C_{\text{boundary}} , \quad (13)$$

where P_{expected} represents expected farm power and the remaining terms represent constraint penalties.

3.3.3 Wind Scenarios

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3.3.4 Genetic Operators

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3.4 OpenFAST Validation

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4 Results

4.1 Surrogate Model Performance

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An example results table is shown in Table 3.

Table 3: Example surrogate model performance.

Model	MAE (dBA)	RMSE (dBA)	R^2
Model A	0.10	0.15	0.999
Model B	0.15	0.22	0.997
Model C	0.20	0.28	0.995

4.2 Genetic Algorithm Performance

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4.3 Optimized Layout

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4.4 OpenFAST Validation

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5 Discussion

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6 Limitations

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Example limitations could be presented as:

- OpenFAST validation currently uses deterministic uniform inflow.
- The aeroacoustic surrogate is valid only within the domain represented by its training data.
- The Genetic Algorithm does not guarantee identification of the global optimum.
- Additional turbulence conditions would require further high-fidelity simulations.

7 Future Work

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8 Conclusion

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References

- [1] Author, *Example OpenFAST Reference*, Publisher or Journal, Year.
- [2] Author, *Example FLORIS Reference*, Publisher or Journal, Year.

A Optimization Parameters

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B Additional Results

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